L2: No Source, No Agreement — A Forensic Protocol for Literature Values in Model-Assisted Physics

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Abstract

We report a self-inflicted error in our own published-preprint series and turn it into a transferable method. A table of experimental ratios for a viscoelastic vortex experiment was labelled *independent literature, outside the model* and compared against two model outputs. The two model outputs landed inside the two quoted intervals. The intervals were later shown to exist in no source: four primary papers were downloaded and scanned in full (PDF text *and* LaTeX source), and the quoted numbers occur nowhere in them. The claim was withdrawn. Two independent forensic tests are introduced and released as code: *repository archaeology*, which asks whether a value ever carried a source record *before* the agreement that uses it was asserted, and *source-existence scanning*, which asks whether the quoted interval exists in the source corpus at all. We show that the two apparent confirmations were algebraically dependent — the second ratio is the square of the first divided by a locked constant — so the apparent two-point agreement rested on a single degree of freedom resting on a value with no source. Of the six literature quantities used in the two affected tests, three could not be traced to any source and two more were numerically wrong. We give a six-step detection protocol, state plainly what the two tests can and cannot decide, and release the tool, its independent self-test (7/7) and the forensic figure.

1 The problem: intervals are the most dangerous artefact

When a model output is compared with a literature value, the two natural checks are *is the number right?* and *does the model agree?*. Neither of them can detect a value that was never in the literature. A point value that is wrong is caught the moment one opens the source; an *interval* that does not exist can only be caught by exhausting the source, which is expensive and, unlike a wrong number, has no error signature.

Three properties make intervals especially prone to this failure.

1. *They are cheap to invent and hard to falsify.* Two round numbers read as a measurement. "$r \in [1.55, 1.70]$" looks exactly like a summary of a figure axis.

2. *They are self-sealing.* If a model value happens to land inside the interval, the agreement appears to confirm the value — and the value appears to confirm the model. No external check is triggered.

3. *They evade the falsification asymmetry.* A wrong point value propagates a wrong prediction. A phantom interval propagates a wrong *confirmation*, which is worse, because it is not visible in any downstream computation.

This letter is a case study of exactly that failure, in our own work, with the forensic method and the released tool.

2 The case

In the preprint series later numbered SS-XIV, test T35 compared the dimensionless ratios predicted for a 100 ppm polyacrylamide-oxide solution in a two-disk tank against two quoted intervals,

$$r_{\mathrm{PEO}}/r_{\mathrm{water}} \in [1.55,\ 1.70], \qquad \Delta h_{\mathrm{PEO}}/\Delta h_{\mathrm{water}} \in [0.30,\ 0.35], \tag{1}$$

labelled *independent literature (model-external)*. The model outputs were

$$r_{\mathrm{PEO}}/r_{\mathrm{water}} = 1.626037936551738, \qquad \Delta h_{\mathrm{PEO}}/\Delta h_{\mathrm{water}} = 0.314025361279003, \tag{2}$$

and Eq. (2) lies inside Eq. (1) in both cases — the intervals were read off as confirmations.

2.1 Provenance of the intervals

- The quoted intervals entered the repository in commit `043f2ee` (19 October 2026, 20:14), already in the *independent literature* frame, in the same commit that asserted the agreement.
- A source audit 56 minutes later (commit `96467dd`, 21:10) downloaded four primary papers and scanned them in full: Groisman & Steinberg (2000) [1]; van Buel & Stark (2022) [2]; Guido et al. (2021) [3]; Burghelea et al. (2006) [4].
- In the LaTeX source of Groisman & Steinberg the strings `1.55`, `1.70`, `0.30`, `0.35`, *funnel*, *dimple*, *core* and *free surface* do not occur at all; the extracted PDF text (19 519 characters) contains neither `1.55` nor `1.70`. That experiment is polyacrylamide plus syrup between two parallel disks, $d/R = 0.263$, and its figure inventory (schematic; $\sigma/\sigma_{\mathrm{lam}}$ versus $\dot{\gamma}$; snapshots; two Fourier spectra) contains no free-surface or dimple measurement of any kind.
- The intervals were therefore withdrawn and the comparison removed.

The underlying physics is real and sourceable — the same literature reports a flow-resistance increase factor $\sigma/\sigma_{\mathrm{lam}} = 12$ and 19 for 10 and 20 mm gaps, a critical Weissenberg number $\mathrm{Wi}_c = 12$, and a boundary-layer thickness as the new length scale of elastic turbulence — but the quoted interval pair is traceable to nothing.

3 The two forensic tests

3.1 Repository archaeology

The first test does not look for the value in the literature at all; it looks for it in the project's own history. A literature value that was transcribed from a source must, at some point, have been

written down *as a literature value* before the model output was compared with it. We therefore compare three first appearances in the commit history:

$$t_{\text{first}}: \text{ first standalone occurrence of the token,} \tag{3}$$

$$t_{\text{ref}}: \text{ first occurrence in a reference frame,} \tag{4}$$

$$t_{\text{claim}}: \text{ first occurrence in an agreement claim.} \tag{5}$$

If $t_{\text{ref}} = t_{\text{claim}}$, the value was born as an agreement: there is no earlier record in which it was a sourced datum, and the label *independent literature* was applied at the moment of the claim rather than derived from a source.

Two implementation details matter and are enforced in the released tool. First, the search must be for the *standalone token*: a substring search for `1.55` finds the unrelated quantity `1.556152e+05` in the project's first commit and would manufacture a false provenance record. Second, the reference and claim frames are detected in a window of ± 3 lines, because a citation and its number are routinely split across two lines.

3.2 Source-existence scanning

The second test asks whether the quoted interval exists in the source corpus at all: every arXiv LaTeX source and every PDF in the corpus is searched for the number with token boundaries, and the number of files in which the members of an interval co-occur is reported separately. The scan must report failures explicitly; a PDF whose text cannot be extracted is counted as an error, never as a zero.

The results for the corpus of 100 files (81 PDFs, of which 17 are the primary papers, and 19 arXiv LaTeX source files) are:

Table 1: **Existence scan of the literature values used in T35 and T36.** Corpus: 100 files (81 PDFs + 19 arXiv LaTeX sources). Matches are counted with token boundaries. *Co-occurrence* counts the files in which all members of an interval appear together.

Value	Repository verdict	Matches	Co-occurrence
`1.55`	GHOST_CANDIDATE	0	—
`1.70`	GHOST_CANDIDATE	0	—
`1.55` + `1.70` (interval)	GHOST_CANDIDATE	0	0
`0.30`	GHOST_CANDIDATE	4	—
`0.35`	HAS_PROVENANCE	5	—
`0.30` + `0.35` (interval)	GHOST_CANDIDATE	9	3
`2.25` (glitch band)	GHOST_CANDIDATE	2	—
`0.0160` (I_{crust}/I)	GHOST_CANDIDATE	0	—
`0.014` (I_{crust}/I, corrected)	GHOST_CANDIDATE	4	—

Table 1 contains the lesson of the whole letter in two lines. The numbers `1.55` and `1.70` occur nowhere, so the interval is phantom. But `0.30` and `0.35` *do* occur, four and five times, and they co-occur in three files — a 511 keV line-modelling paper, an antihydrogen hyperfine paper and two INTEGRAL flux maps — in contexts with nothing to do with viscoelastic vortices. *Token presence is not provenance, and co-occurrence without context is not provenance either.* Only the semantic reading of the source settles it, which is precisely why the cheapest automatable test — the archaeology — has to be run first, while the claim is still cheap to withdraw.

3.3 The two apparent confirmations were one

The two agreements in Eq. (1) are algebraically dependent. With $a \equiv \mathcal{C}_{\mathrm{univ}}$ and the locked μ_s,

$$\frac{\Delta h_{\mathrm{PEO}}}{\Delta h_{\mathrm{water}}} = \frac{1}{a^2(1+\mu_s)} = \left(\frac{u_{\max}^{\mathrm{PEO}}}{u_{\max}^{\mathrm{water}}}\right)^2 \frac{1}{1+\mu_s}, \tag{6}$$

because the free-surface depression scales as the inverse square of the Burgers core radius, $r_c \propto a$. Numerically the identity in Eq. (6) holds to 5.6×10^{-17}, and $r_{\mathrm{PEO}}/r_{\mathrm{water}} = \mathcal{C}_{\mathrm{univ}}$ to 2.2×10^{-16}. The second interval agreement is therefore a reparametrisation of the first, not an independent test of it: the pair rests on a single degree of freedom. Combined with the fact that the intervals were phantom, the apparent two-point confirmation collapses to zero.

4 The protocol

Six steps, of which only the first three are new practice for most authors.

1. **Quote a source line for every literature value.** A file, a table or figure number, and a page. "Literature" without a locatable location is not a citation.

2. **Ask whether the source *contains* it.** The question is existence, not equality. For an interval, ask whether both members appear together, in the same context.

3. **Scan the machine-readable source.** For arXiv papers, the LaTeX source finds figure captions, table notes and supplementary material that a PDF text extraction misses, and it makes the search reproducible.

4. **Recover the first appearance.** A version-controlled project keeps a time-stamped record of when each value entered and in which frame. The search must respect token boundaries.

5. **Test the ordering.** A value called independent must have been recorded as sourced *before* the agreement that uses it. If the first reference and the first claim are the same commit, the independence was asserted, not derived.

6. **Withdraw rather than repair.** When the number cannot be traced, the claim citing it is withdrawn even if the physics survives. Replace it with the value that *is* sourceable, and record the withdrawal.

Step 6 is the expensive one and the reason step 1 is cheap: a withdrawal costs a paragraph, whereas an untraceable confirmation costs the reader the ability to trust the rest of the table.

5 What the two tests can and cannot decide

We state the limits explicitly, because a forensic tool that overstates its reach becomes the next ghost.

- *The archaeology test says nothing about truth.* It reports whether a value carried a source record in *this* repository before it was used in a claim. The corrected crust-fraction value

$\Delta I/I \geq 0.014$ is genuinely sourced (it occurs in the LaTeX source of Link et al. [5]) and is nevertheless flagged, because the repository had never transcribed that source before the audit. A flag is a request for a source line, not a refutation.

- *The repository cannot establish origin.* It establishes only that no provenance record exists. Whether the interval was misremembered, carried over from an earlier working note, suggested by a language model during drafting, or produced by a slip cannot be determined from the repository, and we do not claim it.

- *The existence scan bounds the printed record, not the physics.* Zero matches means the quoted numbers appear nowhere in the scanned corpus; it does not prove that no measurement resembling them was ever made. What it does prove is that the claim as written had no retrievable basis.

6 The scale of the problem in one test pair

The audit of tests T35 and T36 used six literature quantities: the two intervals, the glitch amplitude $(2.25 \pm 0.35) \times 10^{-6}$, the healing fraction $Q \in [0.05, 0.18]$, the crust-fraction ratio $I_{\rm crust}/I$, and the glitch relaxation timescales. Three could not be traced to any source; two more were numerically wrong and were replaced by measured values ($I_{\rm crust}/I = 0.0160 \rightarrow \geq 0.014$ from Link et al., and arbitrary 6- and 60-day relaxation times $\rightarrow$ measured 1.2 min, 0.53, 3.29 and 19.07 days from Dodson et al. [6]); only the sixth was already correct. **Five of six required correction.** In this series that ratio improved to zero only because the sources were eventually downloaded and read.

7 Reproducibility

The complete forensic machinery is released with this letter:

- `ghost_audit.py` — the two tests: repository archaeology (first appearance, reference frame, claim frame, with token-boundary enforcement and a ±3-line context window) and source-existence scanning (with explicit co-occurrence counting and explicit failure reporting).

- `mini_check_l2.py` — an independent self-test that builds synthetic repositories and corpora and checks the four verdicts (`GHOST_CANDIDATE`, `HAS_PROVENANCE`, `NOT_A_LIT_VALUE`, `ABSENT`), the token-boundary rule, the co-occurrence count and the real case: 7/7 pass.

- Figure 1 — the life cycle of the phantom interval, the existence scan, the algebraic collapse of the two apparent confirmations, and the protocol.

The tool does not import the solver and does not depend on the physical model; it operates on a repository and a corpus. Execution instructions are given in the accompanying repository.

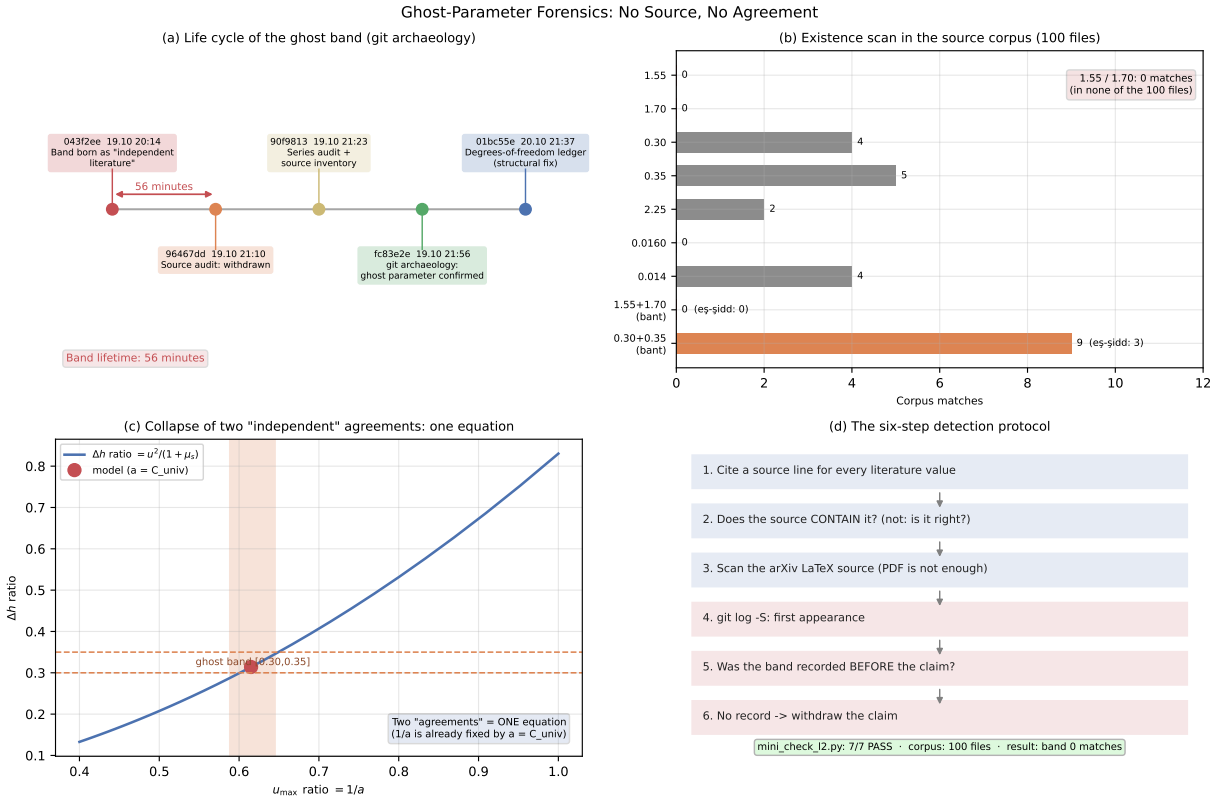


Figure 1: **Forensic summary of the phantom interval.** (a) The interval was created as *independent literature* in commit `043f2ee` and withdrawn 56 minutes later by the source audit; the archaeology confirmation and the structural fix (the degrees-of-freedom ledger) follow. (b) Existence scan of the source corpus: `1.55` and `1.70` occur nowhere; `0.30` and `0.35` occur, and co-occur in three files, but in unrelated contexts. (c) The two apparent confirmations are one equation: the depression ratio is the square of the velocity ratio divided by $1+\mu_s$. (d) The six-step protocol.

References

[1] Groisman, A. A. & Steinberg, V. (2000). Elastic turbulence in a polymer solution. *Nature* ***405***, 53–61. arXiv:nlin/0104052.

[2] van Buel, J. & Stark, P. D. (2022). Elastic turbulence in dilute polymer solutions. arXiv:2202.12714.

[3] Guido, S. *et al.* (2021). Development of vortex based on polymeric solution. arXiv:2110.03004.

[4] Burghelea, A., Segre, D. & Steinberg, V. (2006). Elastic turbulence in a dilute polymer solution. arXiv:physics/0609112.

[5] Link, F., Epstein, D. L. & Lattimer, J. M. (1999). Glitching pulsars. *Phys. Rev. Lett.* ***83***, 3807–3810.

[6] Dodson, R. N. *et al.* (2002). Frequency behaviour of the Vela pulsar during the glitch of 2000 January. *Astrophys. J.* ***574***, 471–475. arXiv:astro-ph/0201005.